Quantitative Portfolio Project

Group 3

April 2026

Project Overview

Data and Period

- Assets: SPY, BND, BIL, and 11 sector ETFs (XLC, XLY, XLP, XLE, XLF, XLV, XLI, XLB, XLRE, XLU, XLK)
- Market proxy: SPY
- Risk-free proxy: BIL
- Backtest period: January 2019 – December 2023

Portfolios Constructed

- Task 1: SPY Buy-and-Hold
- Task 2: 60/40 Stock-Bond Portfolio
- Task 3: Fast Algorithm (Single-Index Markowitz-Sharpe Portfolio)
- Task 4: Custom Momentum Strategy

Evaluation Metrics

- Return, Standard Deviation, VaR, Sharpe Ratio
- CAPM: Beta and Jensen's Alpha
- Drawdown and rolling return histograms

Task 1: SPY Buy-and-Hold Strategy

Description and Assumptions

- Passive benchmark strategy fully invested in SPY
- No rebalancing
- No leverage, no short selling, no transaction costs
- Risk-free rate proxied by BIL

Performance Highlights

- Cumulative Return: 106.37%
- Annualized Return: 15.62%
- Annualized Volatility: 20.99%
- Sharpe Ratio: 0.72
- Beta: 1.00, Jensen's Alpha: 0.00%

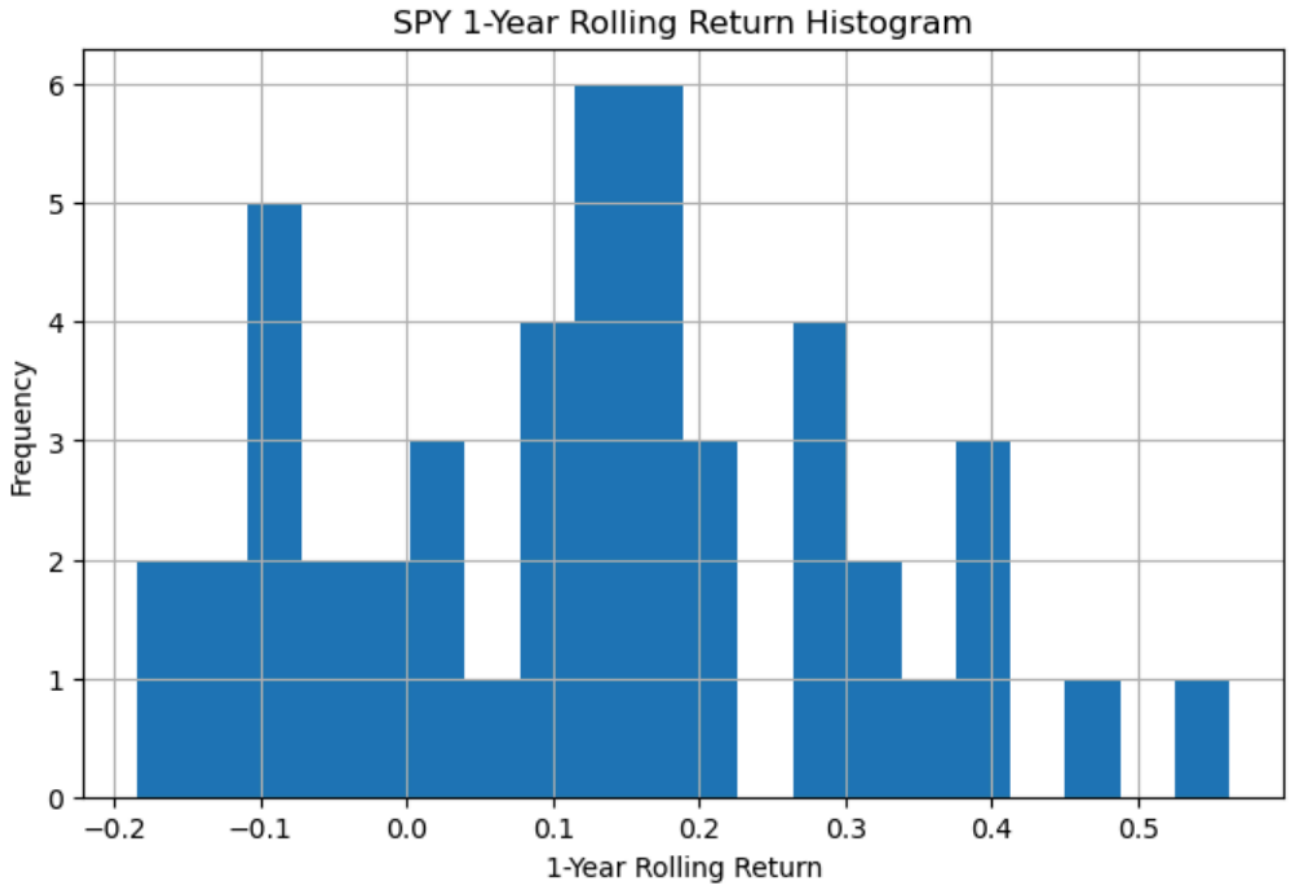
Interpretation

- Strong long-term growth
- High volatility and large drawdowns
- Serves as the core benchmark for all other strategies

Task 1: Performance Summary

Metric	Value
Annualized Return	15.62%
Annualized Volatility	20.99%
Sharpe Ratio	0.72
Maximum Drawdown	-33.72%
VaR (95%)	-1.93%
CVaR (95%)	-3.22%
Beta vs SPY	1.00
Jensen's Alpha	0.00%

Task 1: Histogram

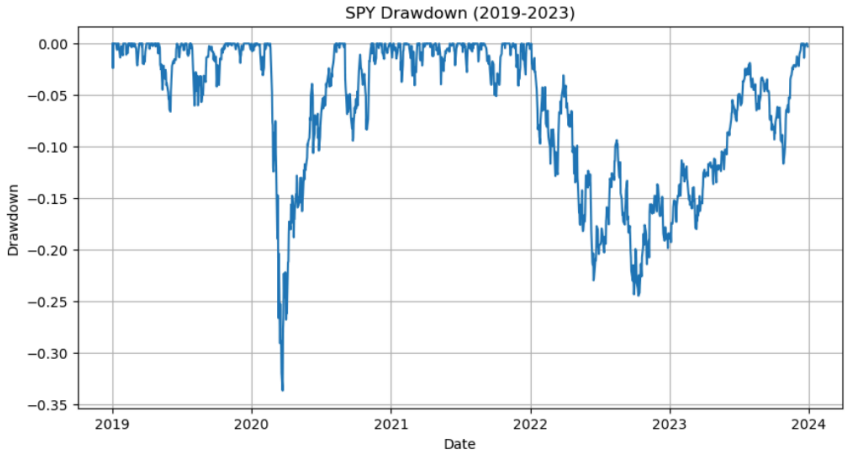


Insight: Most rolling 1-year returns fall between 10% and 30%, with negative observations during crisis periods and strong right-tail outcomes during recoveries.

Task 1: NAV and Drawdown



(a) NAV



(b) Drawdown

Insight: NAV rises strongly over the sample, but the 2020 COVID shock and the 2022 tightening cycle lead to large interim losses.

Task 2: Stock-Bond Portfolio (60/40)

Description and Assumptions

- 60% SPY and 40% BND
- Long-only, fully invested
- Risk-free rate proxied by BIL

Rebalance Description

- Quarterly rebalancing
- Weights reset to target proportions at each quarter start
- Between rebalances, weights drift with market returns

Economic Role

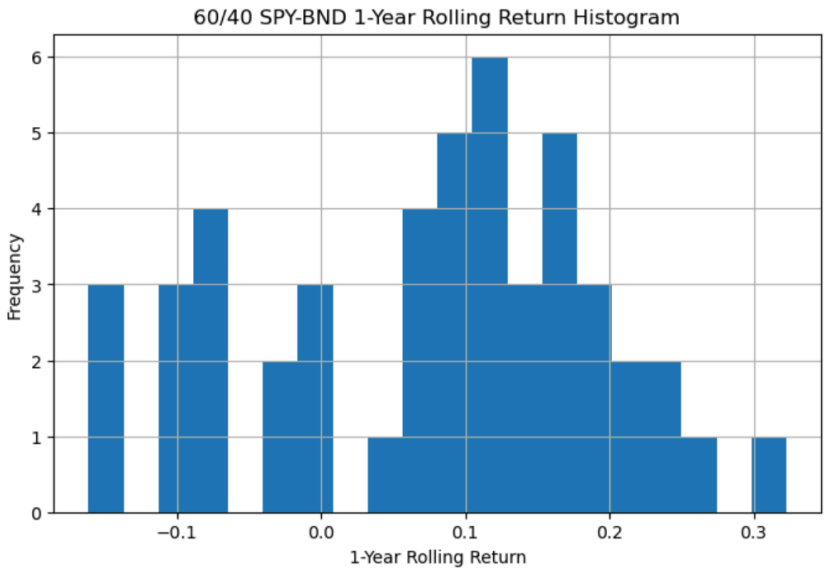
- Equity for growth
- Bonds for stabilization and lower downside risk

Task 2: Performance Summary

Metric	Value
Annualized Return	10.02%
Annualized Volatility	13.15%
Sharpe Ratio	0.66
Maximum Drawdown	-21.03%
VaR (95%)	-1.17%
CVaR (95%)	-1.99%
Beta vs SPY	0.61
Jensen's Alpha	-0.48%

Interpretation: Lower return than SPY, but much lower volatility, smaller drawdown, and better downside protection.

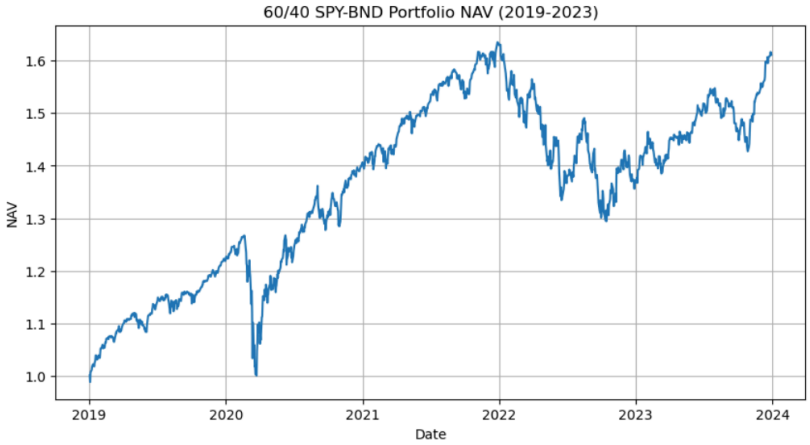
Task 2: Histogram



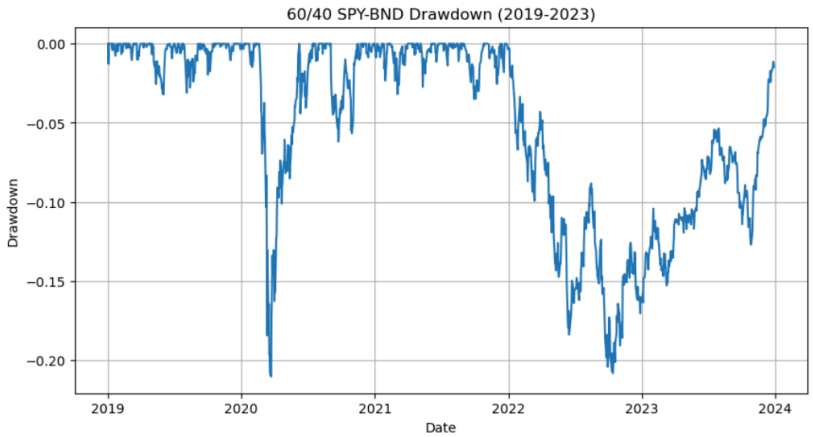
Interpretation

- Most observations are concentrated between 5% and 20%, indicating stable performance. The distribution is narrower than SPY, reflecting lower volatility.
- The left tail is relatively mild, suggesting reduced downside risk.

Task 2: NAV and Drawdown



(a) NAV



(b) Drawdown

Insight: The 60/40 strategy delivers smoother growth and a more concentrated return distribution, consistent with lower risk and reduced variability.

Task 3: Fast Algorithm (Single-Index Markowitz-Sharpe)

Description and Assumptions

- Uses Sharpe single-index model and Elton-Gruber-Padberg Fast Algorithm
- Avoids full covariance matrix estimation
- Universe: 11 sector ETFs
- Long-only, fully invested, no leverage

Model Framework

$$R_i - R_f = \alpha_i + \beta_i(R_M - R_f) + \varepsilon_i$$

Key Inputs

- Expected excess return
- Beta with respect to SPY
- Residual variance

Task 3: Parameter Determination and Fast Algorithm

Parameter Determination

- Rolling estimation window: 252 trading days
- For each sector ETF:

$$\bar{r}_i = E[R_i - R_f]$$

$$\beta_i = \frac{\text{Cov}(R_i - R_f,\ R_M - R_f)}{\text{Var}(R_M - R_f)}$$

$$\sigma^2_{\varepsilon,i} = \text{Var}\left[(R_i - R_f) - \alpha_i - \beta_i(R_M - R_f)\right]$$

Fast Algorithm Construction

- Rank sectors by excess-return-to-beta ratio:

$$\frac{\bar{r}_i}{\beta_i}$$

- Determine cut-off value C^*
- Include only assets above the cut-off
- Compute weights analytically

Task 3: Rebalance, Beta, and Weight Characteristics

Rebalance Mechanism

- Quarterly rebalancing
- At each quarter start:
  1. Collect latest 252 trading days
  2. Estimate parameters
  3. Rank sectors
  4. Determine cut-off
  5. Update weights

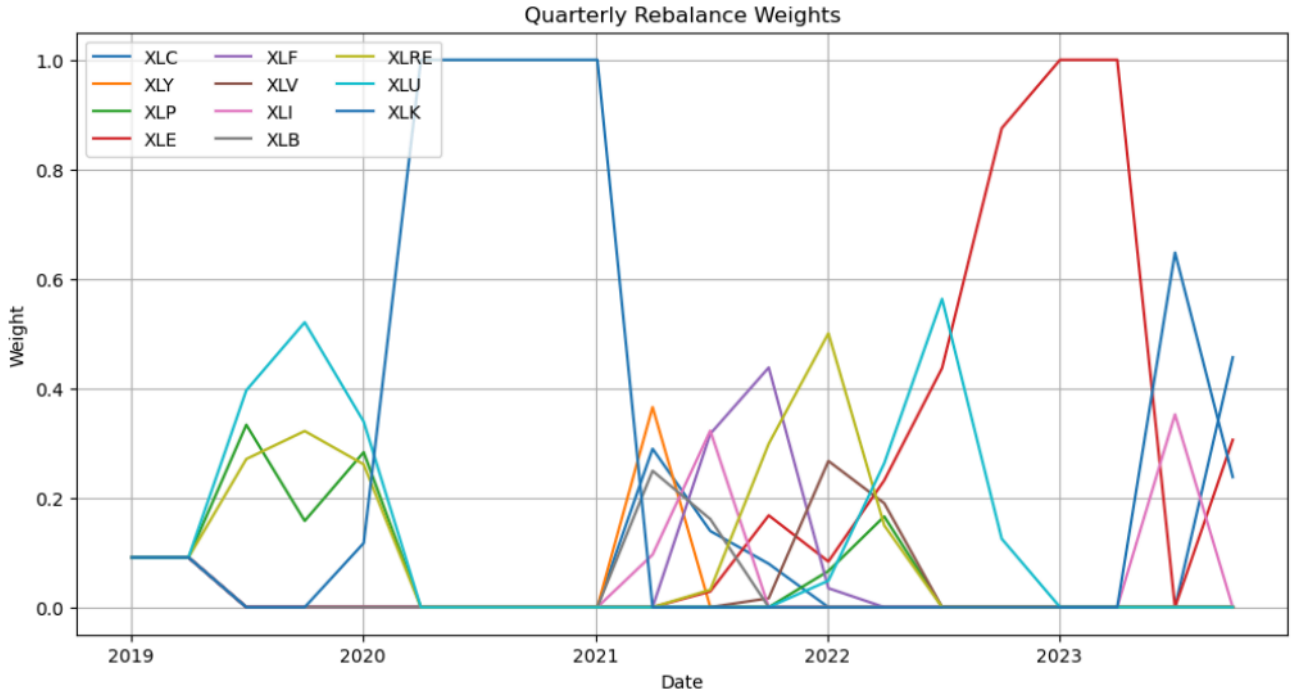
Beta Interpretation

- Realized portfolio beta: 0.92
- Slightly lower market exposure than SPY
- Positive alpha indicates return beyond market exposure

Weight Characteristics

- Dynamic and sometimes highly concentrated (See next slide)
- Strong concentration in technology during 2020–2021
- Strong concentration in energy during 2022–2023

Task 3: Weight Dynamics



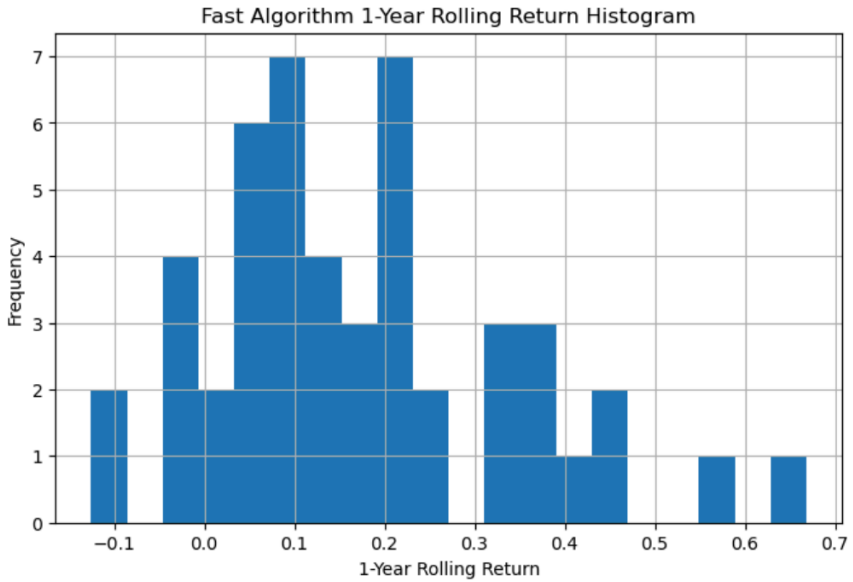
Insight: The analytical fast algorithm can produce concentrated allocations when a few sectors dominate in risk-adjusted attractiveness.

Task 3: Performance Summary

Metric	Value
Annualized Return	17.29%
Annualized Volatility	23.52%
Sharpe Ratio	0.72
Maximum Drawdown	-32.81%
VaR (95%)	-2.21%
CVaR (95%)	-3.48%
Beta vs SPY	0.92
Jensen's Alpha	3.25%

Interpretation: The Fast Algorithm outperforms SPY and 60/40 in return and generates positive alpha, but concentration raises volatility and tail risk.

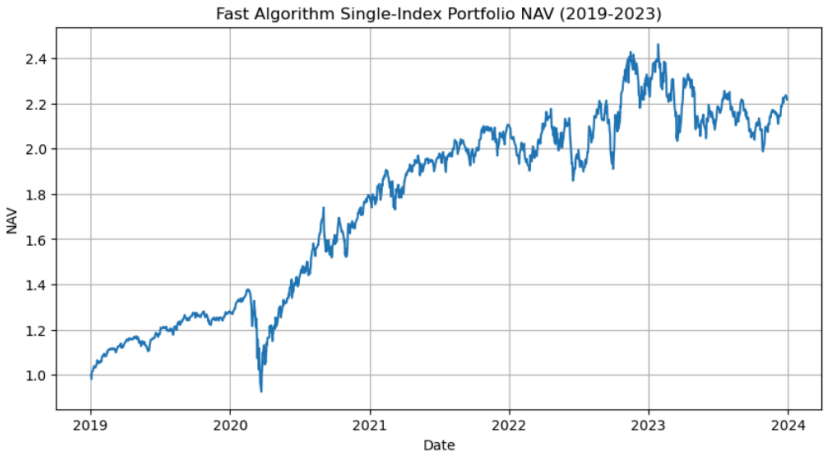
Task 3: Histogram



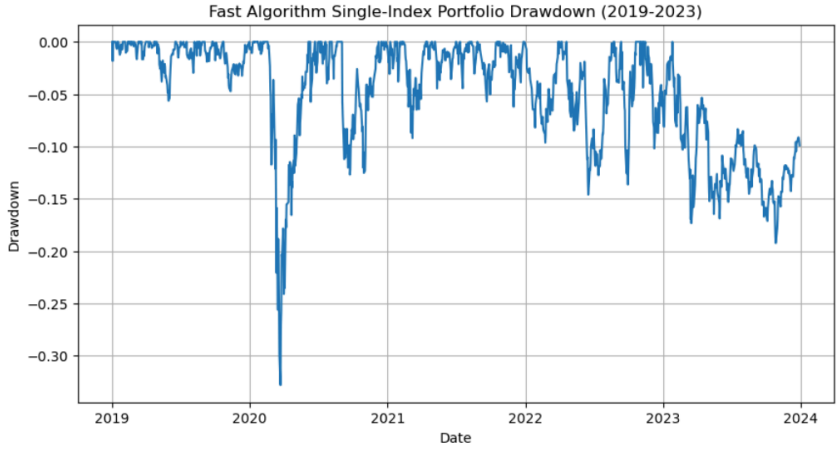
Interpretation

- A pronounced right tail is observed, with several periods achieving very high returns. The left tail is also present, suggesting increased downside risk compared to diversified strategies.
- Overall, the strategy exhibits higher return potential at the cost of greater variability and tail risk.

Task 3: NAV and Drawdown



(a) NAV



(b) Drawdown

Insight: The strategy exhibits strong long-run growth and a relatively large max drawdown, reflecting both upside potential and concentration-driven variability.

Task 4: Custom Momentum Strategy

Description and Assumptions

- Dynamic sector allocation based on recent performance trends
- Objective: capture momentum effects by overweighting strong sectors
- Long-only, fully invested, monthly rebalancing

Indicator Definition

$$\text{Momentum}_i = \frac{P_t}{P_{t-L}} - 1$$

where P_t is the current price and L is the lookback window.

Interpretation

- Higher momentum indicates stronger recent trend
- Sectors with stronger trends receive larger portfolio weights

Task 4: Construction, Rebalance, and Economic Intuition

Portfolio Construction

- Rank sector ETFs by momentum scores
- Select top-performing sectors
- Normalize weights to sum to one

Rebalance Mechanism

- Monthly rebalancing
- At each month start:
  1. Compute momentum indicator
  2. Rank sectors
  3. Select strongest signals
  4. Assign normalized weights

Economic Intuition

- Based on momentum effect
- Strong past performers tend to continue outperforming
- Strategy captures sector rotation over time

Task 4: Weight Dynamics



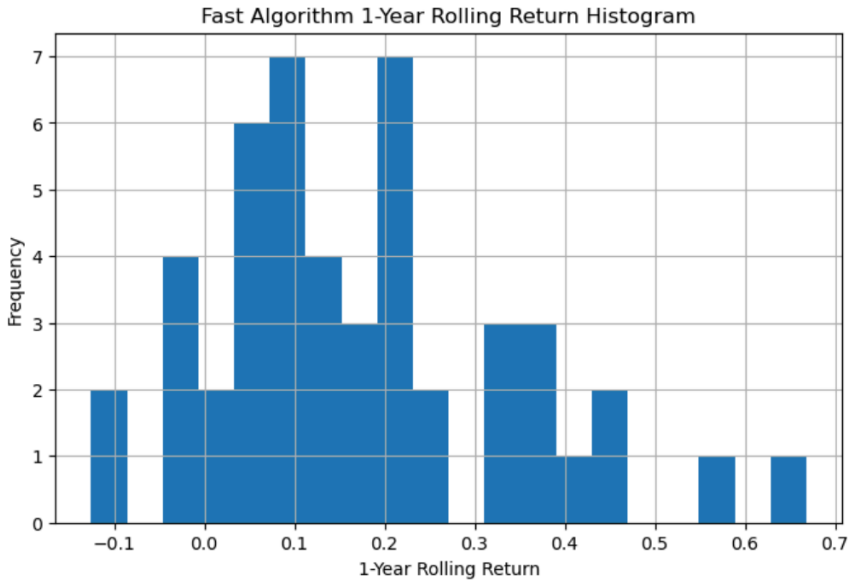
Insight: The strategy shows pronounced sector concentration during strong momentum periods, particularly in energy during 2022–2023, highlighting both its return potential and its sensitivity to sector-specific risk.

Task 4: Performance Summary

Metric	Value
Annualized Return	18.04%
Annualized Volatility	22.35%
Sharpe Ratio	0.78
Maximum Drawdown	-30.32%
VaR (95%)	-2.01%
CVaR (95%)	-3.28%
Beta vs SPY	0.88
Jensen's Alpha	4.21%

Interpretation: This is the strongest strategy overall, with the highest return, highest Sharpe ratio, and highest alpha.

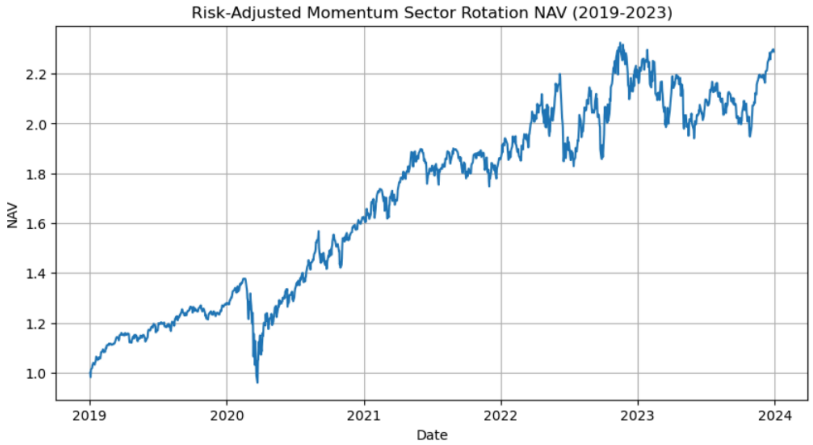
Task 4: Histogram



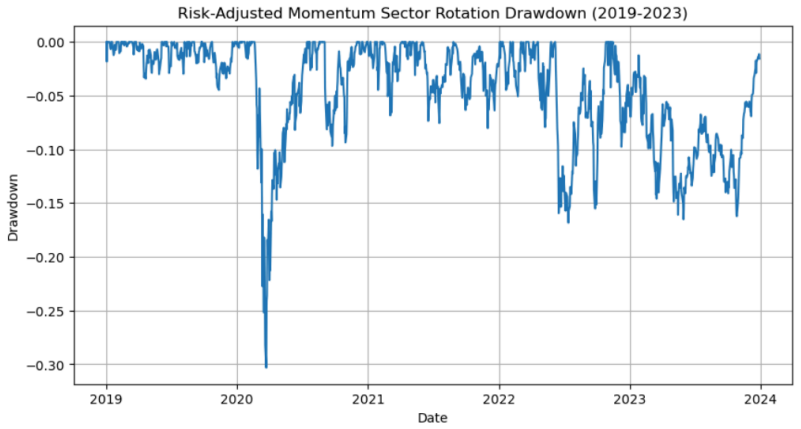
Interpretation

- Most returns are concentrated in the positive range, indicating consistent outperformance over time. However, compared to more diversified strategies, the spread is wider, reflecting higher variability.
- Overall, the strategy delivers strong upside potential while maintaining moderate downside risk.

Task 4: NAV and Drawdown



(a) NAV



(b) Drawdown

Insight: The custom strategy shows consistent outperformance, improved downside control relative to Fast Algorithm, and a favorable right-tailed return distribution.

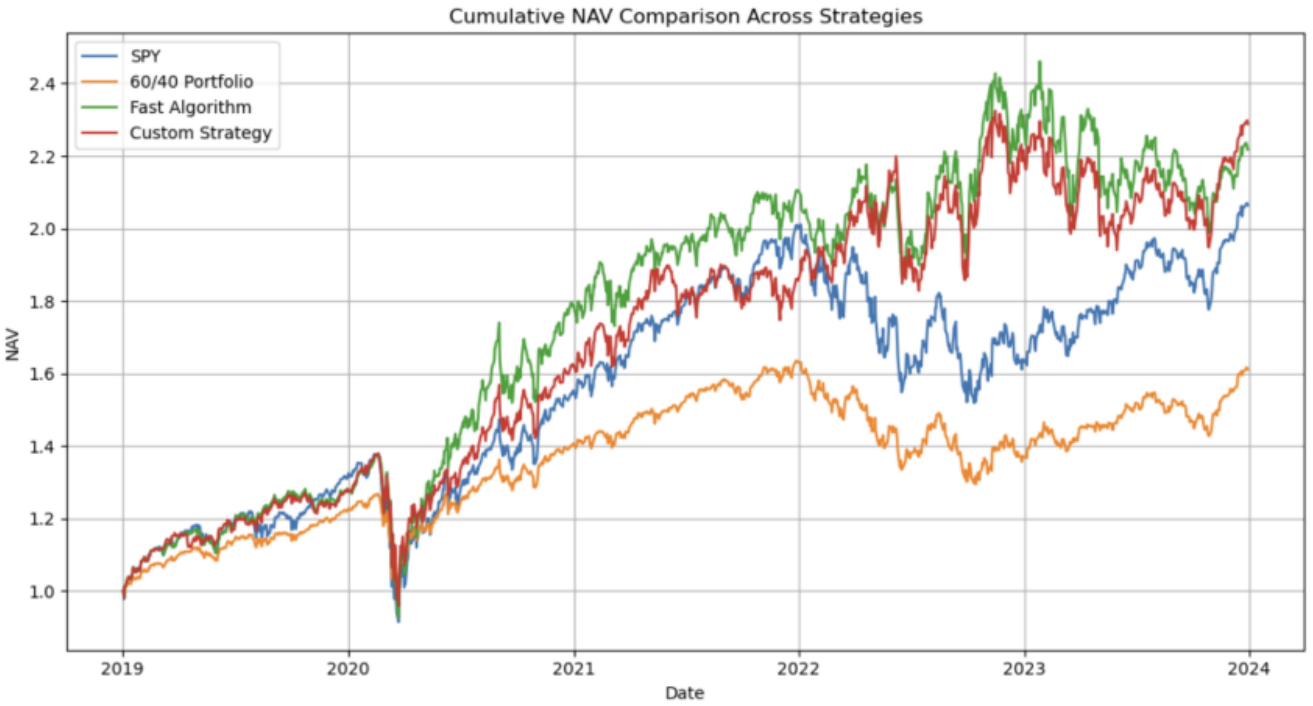
Task 5: Overall Comparison

Strategy	Return	SD	VaR (95%)	Sharpe	MDD
SPY	15.62%	20.99%	-1.93%	0.72	-33.72%
60/40	10.02%	13.15%	-1.17%	0.66	-21.03%
Fast Algo	17.29%	23.52%	-2.21%	0.72	-32.81%
Custom	18.04%	22.35%	-2.01%	0.78	-30.32%

Task5: Key Comparison

- Custom strategy: best overall performance
- 60/40: lowest volatility and smallest drawdown
- Fast Algorithm: strong alpha, but higher concentration risk
- SPY: benchmark with pure market exposure

Task 5: Graphical Comparison



Insight: The custom strategy slightly outperforms the Fast Algorithm, while 60/40 provides the smoothest but lowest-return path.

Task 5: Final Remarks

- Passive benchmark provides a useful baseline but no alpha
- 60/40 improves stability and reduces downside risk
- Fast Algorithm demonstrates the power of the single-index framework
- Custom momentum strategy delivers the best return and risk-adjusted performance

Conclusion: Active strategies can outperform passive benchmarks when appropriate signals are used, but higher return often comes with greater concentration and model complexity.